

Gaurav Kumar

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Education

University of South Carolina

Masters in Computer Science — GPA: 4.0 / 4.0

Courses: Analysis of Algorithms, Compiler, Computer Architecture, Network System Security

Columbia, SC

01/2024 – 05/2025

Indian Institute of Technology Roorkee

Bachelors in Engineering Physics — GPA: 7.03 / 10

Courses: Data Structures and Algorithms, Optimization Theory, Statistics, Linear Algebra

Roorkee, India

07/2016 – 05/2020

Research Experience

Sutton Lab

Graduate Research Assistant | Python, HPC, Rdkit, ASE

Columbia, SC

01/2024 – Present

- Working on leveraging generative diffusion models to propose molecular structures, addressing combinatorial scaling challenges and enhancing material structure prediction efficiency.
- Developed tree-based models to accurately predict JV Reverse Scan PCE of Perovskite Materials and worked on fine-tuning encoder based model on molecular structures for precise band gap predictions.

Work Experience

Yojak | Startup

Software Engineer | Django, React, Kafka, Docker, Redis, Elasticsearch

Gurgaon, India

05/2022 – 12/2023

- Designed and developed a React Native cross-platform e-commerce app with QR SKU scanning and real-time stock updates, boosting user engagement by 35% and reducing order processing time while ensuring transaction fault tolerance.
- Architected and implemented an integrated ERP system in an agile environment, consolidating order processing, accounting, and supply chain management. Integrated Apache Kafka to enable an event-driven architecture for asynchronous communication. Automated 15+ manual processes, saving over 40 hours weekly and enhancing operational efficiency.
- Led the creation of automated deployment pipelines using AWS (CloudFront, S3) and Docker for backend and front-end applications. Implemented caching strategies that reduced deployment time by 50%.

Fractal Analytics

Data Scientist | Python, SQL, Tableau, Jenkins, AWS SageMaker, Git

Mumbai, India

10/2020 – 05/2022

- Built a Customer Lifetime Value (CLV) model for a retail client, optimizing marketing spend across multiple channels and enhancing ROI by 1%, resulting in a \$10M increase in annual revenue.
- Developed and deployed an implicit collaborative filtering recommendation model and engineered a seasonal prioritization multiplier, resulting in a 7% (\$100k/month) increase in sales validated through A/B testing.
- Increased lead conversion rate by 20% by developing a lead scoring model using tree-based algorithms and deploying it in real-time on Kubernetes with FastAPI.

Technical Skills

Languages: Python, C++, Rust, Golang, Java, JavaScript, TypeScript

Frameworks: Django, Django Rest Framework/, React, React Native, REST API, Node.js, Express, Next.js

Databases: MySQL, PostgreSQL, MongoDB, Redis, GraphQL, Prometheus, Dynamo DB

Generative AI: GANs, VAE, Diffusion Models, LLMs, Llama, Anthropic, OpenAI, RAG

Cloud Platforms: GCP, AWS (EC2, S3, SQS, Lambda, Cloudwatch, ECR), Azure

DevOps: Linux, Docker, Git, GitHub, Kubernetes, Grafana, Kibana, Logstash, Kafka, Postman, Firebase

Projects

DragAI | *Python, LlamaIndex, Langchain, Hugging Face*

01/2025

- Developed an AI Knowledge Assistant that helps build RAG Applications by crawling entire websites or specific pages from URLs and storing the extracted data in vector or relational databases. The system supports multi-URL fetching, efficient indexing, and leverages a custom LLM to answer queries based on the retrieved information.

Honors and Awards

5th Big Data Health Science Student Case Competition

01/2024

- Winner of 5th National Big Data Health Science Student Case Competition for developing a model predicting sleep stages accurately.